COURSE OVERVIEW

MGMT 648 · Unit 1

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Course topics

Capital budgeting

Free cash flow, NPV, IRR, incremental cash flows, multiple alternatives, real options, and simulation. Sept 12

Company valuation

Two-stage DCF model (Sept 17) and multiples of comparables (Oct 1)

Rates and portfolios

Mean-variance optimization, the cost of capital, bond pricing and duration, immunization. Oct 10

Wrap-Up

Live in-class Q&A. Oct 15

Excel

Still the right tool for a tabular model that adds and subtracts, and the one your colleagues can open, change, and argue with.

AI

Everything else: fetching filings, estimating a beta, optimizing a portfolio, running ten thousand simulations. It also builds the workbook.

Tools

Course materials

Everything is posted at mgmt648.kerryback.com.

Schedule

The home page. Slides and files distributed here.

Syllabus

Grading, meeting times, and what to prepare. Nothing outside class is collected.

Free readings

Welch's *Corporate Finance*, Damodaran's materials

EXCEL

1. Every assumption gets its own labeled cell, in one block at the top
2. Formulas reference those cells, never a number typed inside the formula
3. The same formula runs across every period of a projection
4. Every hardcoded number carries its source beside it

> Write `=B5*(1+$B$6)`, not `=B5*1.05`. Source format: document, date, specific reference — “Company 10-K, FY2025, page 45, revenue note.”

Best practices — structure

Best practices — presentation

CONVENTION	RULE
Blue text	Hardcoded inputs — the cells a reader is meant to change
Black text	Every formula and calculation
Green text	Links to another sheet in the same workbook
Currency	`$#,##0`, with units in the header: Revenue ($mm)
Percentages	One decimal. Valuation multiples as `0.0x`
Negatives	Parentheses, (123), not −123

Zero formula errors on delivery: no `#REF!`, `#DIV/0!`, `#VALUE!`, or `#NAME?`.

Why the colors matter

The color convention is not decoration. It is how a reader knows, at a glance, what they are allowed to change.

For your reader

Blue says “this is an assumption, argue with it.” Black says “this follows from the assumptions, do not touch it.”

AI

1. Go to lab648.kerryback.com
2. Username is your netid-648 (example: abc01-648)
3. Password is your student number, `S0...`

> A Linux container on a cloud server, billed to the school. Security on it is minimal — keep nothing personal there.

The lab

What is running there

The harness

Claude Code, connected to GLM 4.7 — a capable and relatively inexpensive model from Z.ai.

The software

Python and its scientific libraries, plus skills for financial data, spreadsheets, and slides.

The screen

File browser and terminal on the left; the prompt window at the foot of the right-hand panel.

Three things that will trip you up

Refresh

The file browser does not notice a new file on its own. Use the refresh button above it.

No Excel in the lab

Office is not installed. Double-clicking a workbook there does nothing.

Download to open

Workbooks come down to your laptop through the three-dots menu, and open in your own Excel.

AI AND CODING

The large language model (LLM) only writes text

The model cannot open a file, run a calculation, or fetch a filing. It reads text and writes text. That is the entire capability.

What makes it useful is the harness around it. The model emits a request to use a tool; the harness runs the tool and reports the result back.

1. The harness sends the prompt, the conversation, and the list of tools
2. The model answers: use tool X, with these arguments
3. The harness runs X and reports the result back
4. The model decides what comes next — another tool, or an answer for you

> On the lab, GLM writes a Python file, runs it at the terminal, reads the output, and revises. Four tool calls before you see anything.

The loop

What that produces

Excel workbooks

Written cell by cell with Python: inputs as numbers, formulas as formulas.

Charts and figures

Any chart style, custom annotations, labels, fonts

Word and PowerPoint

Real files with editable text, not pictures of them.

Data analysis

Data fetched, merged, filtered, aggregated — statistics and regressions

Advanced data analysis

Simulations, optimizations, machine learning

Web apps

A page someone else can use, from a description of what it should do.

Try this prompt:

> Build an Excel workbook containing a mortgage amortization table. Put the loan amount, rate, and term in labeled input cells at the top. > > Include a chart of the amount going to interest and the amount going to principal each month.

Example

THE WORKBOOK

Inputs in labeled cells, an outputs block below them, 360 rows of schedule.

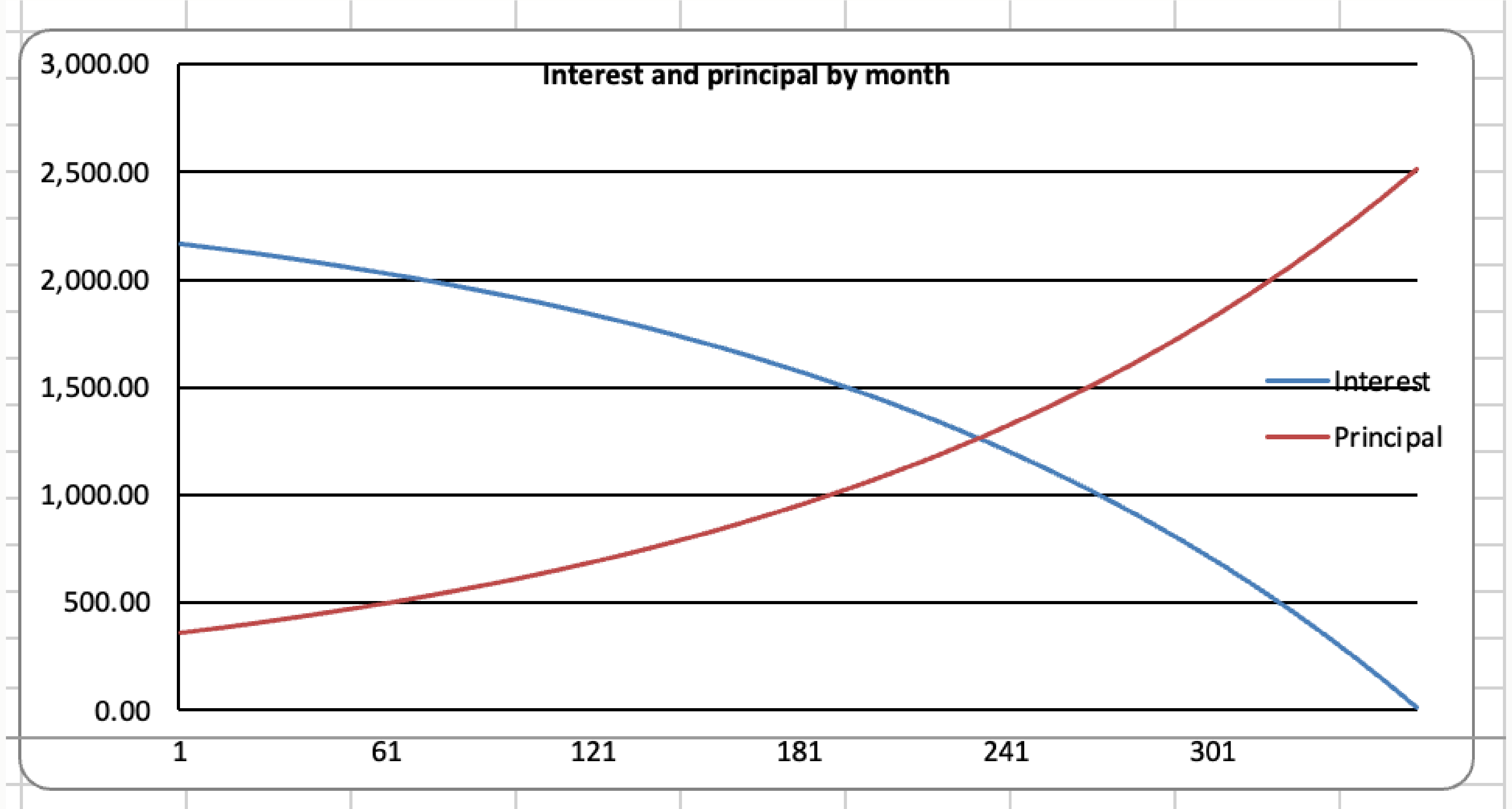
The selected payment cell holds `=$B$12`, not 2,528.27.

Change the rate and the whole sheet recalculates.

C18 =$B$12

	A	B	C	D
1	**Mortgage Amortization**			
2				
3	**Inputs**			
4	Loan amount	**$400,000**		
5	Annual rate	**6.50%**		
6	Term (years)	**30**		
7	Payments per year	**12**		
8				
9	**Outputs**			
10	Periodic rate	0.5417%		
11	Total payments	360		
12	Monthly payment	**$2,528.27**		
13	Total interest	$510,178		
14				
15				
16	**Amortization schedule**			
17	**Month**	**Beginning balance**	**Payment**	**Interest**
18	1	400,000.00	2,528.27	2,166.
19	2	399,638.39	2,528.27	2,164.
20	3	399,274.83	2,528.27	2,162.
21	4	398,909.30	2,528.27	2,160.
22	5	398,541.78	2,528.27	2,158.
23	6	398,172.28	2,528.27	2,156.
24	7	397,800.77	2,528.27	2,154.
25	8	397,427.26	2,528.27	2,152.
26	9	397,051.71	2,528.27	2,150.
27	10	396,674.14	2,528.27	2,148.
28	11	396,294.52	2,528.27	2,146.

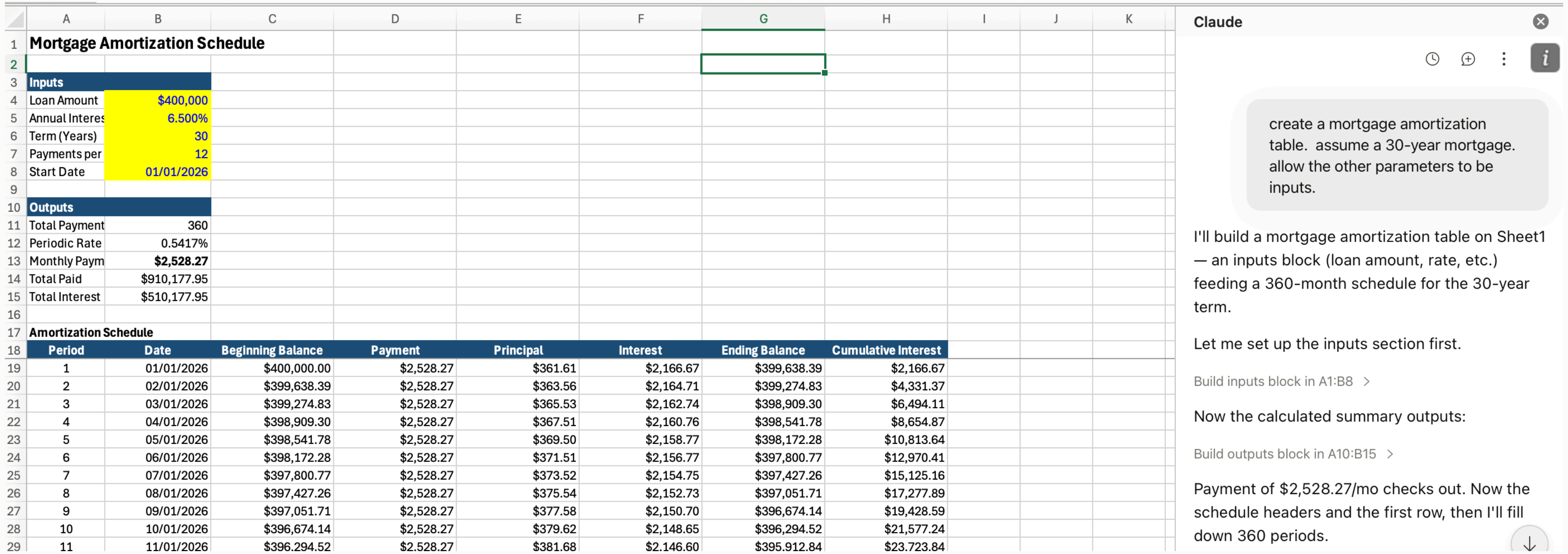
And the chart it asked for



A chart object bound to the cells, not a picture pasted in. Change the rate and the crossover moves with it.

Claude inside Excel

There are also Claude, ChatGPT, and CoPilot plugins for Excel , from `Home → Add-ins`. The chat window sits inside the workbook.



Mortgage Amortization Schedule

Inputs	
Loan Amount	$400,000
Annual Interes	6.500%
Term (Years)	30
Payments per	12
Start Date	01/01/2026

Outputs	
Total Payment	360
Periodic Rate	0.5417%
Monthly Paym	**$2,528.27**
Total Paid	$910,177.95
Total Interest	$510,177.95

Amortization Schedule

Period	Date	Beginning Balance	Payment	Principal	Interest	Ending Balance	Cumulative Interest
1	01/01/2026	$400,000.00	$2,528.27	$361.61	$2,166.67	$399,638.39	$2,166.67
2	02/01/2026	$399,638.39	$2,528.27	$363.56	$2,164.71	$399,274.83	$4,331.37
3	03/01/2026	$399,274.83	$2,528.27	$365.53	$2,162.74	$398,909.30	$6,494.11
4	04/01/2026	$398,909.30	$2,528.27	$367.51	$2,160.76	$398,541.78	$8,654.87
5	05/01/2026	$398,541.78	$2,528.27	$369.50	$2,158.77	$398,172.28	$10,813.64
6	06/01/2026	$398,172.28	$2,528.27	$371.51	$2,156.77	$397,800.77	$12,970.41
7	07/01/2026	$397,800.77	$2,528.27	$373.52	$2,154.75	$397,427.26	$15,125.16
8	08/01/2026	$397,427.26	$2,528.27	$375.54	$2,152.73	$397,051.71	$17,277.89
9	09/01/2026	$397,051.71	$2,528.27	$377.58	$2,150.70	$396,674.14	$19,428.59
10	10/01/2026	$396,674.14	$2,528.27	$379.62	$2,148.65	$396,294.52	$21,577.24
11	11/01/2026	$396,294.52	$2,528.27	$381.68	$2,146.60	$395,912.84	$23,723.84

They need paid accounts or particular types of license agreements. We will not use them.

BEST PRACTICES FOR USING AI

Plan before you act

For any complex operation, ask AI to plan first, or ask it what the best way is to do something, before it begins to work.

Planning pulls the knowledge in

Plan how you will do this before you start. The plan puts the relevant considerations into the conversation, and it answers in light of those rather than from base rates.

Planning breaks the work up

Steps you can check one at a time. Failures stay small and local instead of surfacing at the end, when everything downstream is already wrong.

The problem

Errors usually come from

- AI misunderstood what you wanted, or
- The task could be done in multiple ways, you did not specify, and AI chose the wrong path

What to say:

- Tell me what you are about to do
- Tell me what you did
- Was there any ambiguity in my instructions? Did you make any choices we did not discuss?

Misunderstandings and bad assumptions

Numbers need code

- AI has gotten better at arithmetic, but it is still not to be trusted at math — it is trained to predict language
- Always ask it to run code if you need a number
- It may not be good at predicting what should follow "what is 32 times 17?" but it is very good at generating code "x = 32 * 17; print(x)"

Ask why your idea is bad, not why it is good

Training on human feedback teaches that agreement earns high ratings. Ask “is this valuation sound?” and it will find reasons that it is.

> Invert it. *What is the strongest argument that this project should be rejected?*

AI is not a good self-critic

Ask AI in the same conversation to evaluate its work. It will review its own reasoning and conclude it is sound.

Start a new conversation or ask for a subagent

A new conversation or a subagent will not see the prior reasoning. It will look at it “with a fresh set of eyes.”

Use independent agents as critics

Hallucinations

- Hallucinations are much less frequent now than a few years ago
- Hallucinating code is really not an issue
- Ask for web search and a source for any fact
- The greatest risk is asking for an obscure fact that it cannot find via a web search

Check the outputs

We don't usually check AI code by reading it. We check small cases where we can verify "by hand" (using AI to help with that too).

And keep what works

Keep code that has been vetted. Tell AI to use it again the next time.

Verify and retain code